

Luis Alfredo Cuamatzi Flores

AI Engineer / Software Developer

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SUMMARY

AI Engineer and Software Developer with experience in mobile applications, backend services, and AI model integration. I have worked on real-world projects combining Android, APIs, automation, data processing, and product-oriented interfaces to connect users, data, and business processes.

EXPERIENCE

Software & AI Developer <i>Net2Alliance</i>	Jun 2025 – Present <i>Mexico City, Mexico</i>
- Full-stack and AI development for telecom and CRM platforms. - Built a FastAPI service for sentiment analysis and automated PDF summary generation from call transcripts. - Developed a responsive WebSocket-based messaging client for WhatsApp with real-time notifications, multimedia support, and agent transfers. - Implemented CRUD operations, role-based access control, audit logging, and reporting modules across admin dashboards. - Participated in UI/UX redesigns with Figma and task tracking with Jira Kanban boards.	
AI & Backend Engineer <i>L4L – 3D Garment Reconstruction for Virtual Try-On</i>	Jan 2025 – Dec 2025 <i>Tlaxcala, Mexico</i>
- Built a distributed pipeline for 3D garment reconstruction from a single image. - Developed a FastAPI orchestrator coordinating image capture, avatar confirmation, and GLB model generation. - Model optimization: adapted SMPLer-X and Sewformer to run on local resource-constrained environments, tuning inference pipelines for single-image input. - Integrated SMPLer-X to extract user body pose and shape from a single photograph. - Local Windows server using mayapy and Qualoth for physical cloth simulation and FBX-to-GLB conversion.	
Android Developer <i>Instituto Politécnico Nacional – PIT</i>	Oct 2024 – Aug 2025 <i>Tlaxcala, Mexico</i>
- Designed and built a native Android app to digitize attendance management for the IPN Institutional Tutoring Program. - QR-based attendance scanning with real-time validation, duplicate detection, and active session control. - Weekly schedule builder with overlap detection for classrooms and tutors. - Offline-first sync with Firestore and CSV export for institutional reporting.	

EDUCATION

B.Eng. in Artificial Intelligence Engineering <i>Instituto Politécnico Nacional (IPN) – UPIIT</i>	Aug 2021 – Jan 2026 <i>Tlaxcala, Mexico</i>
Leadership Formation Program <i>Queen Mary University of London</i>	Aug 2024 <i>London, UK</i>

PROJECTS

Lexfy <i>Android + AI project</i>	Dec 2024
- Android app with OCR and AI image generation through a chat interface. - Android client communicating via HTTP with a local Python/Flask backend running EasyOCR, GOT-OCR2_0, and Together AI. Tech: Kotlin, Jetpack Compose, Python, Flask, Firebase, EasyOCR, GOT-OCR2_0, Transformers.	
SmartEMG Vision <i>Android prototype for assistive interaction</i>	Nov 2024
- Mobile prototype combining EMG signal classification and object detection to suggest context-aware actions. - Android client with Jetpack Compose and CameraX sending data via HTTP to local Python servers running YOLOv8 and TensorFlow/Keras.	

- **Tech:** Kotlin, Jetpack Compose, CameraX, Python, Flask, TensorFlow, Keras, YOLOv8.

SmartCrops

Aug 2024

Mobile prototype for greenhouse monitoring

- Prototype for crop monitoring, greenhouse variable simulation, and AI-based chatbot assistance (TomaBot, powered by Gemini).
- Single-activity Jetpack Compose app with CameraX, OkHttp, and Coil.
- **Tech:** Kotlin, Jetpack Compose, Gemini API, CameraX, OkHttp, Coil.

EduStream API

Jun 2023

Web-based educational content management platform

- Web platform for uploading, categorizing, and streaming educational videos by subject.
- MVC backend with multimedia file management and synchronization between physical storage and the database.
- **Tech:** Node.js, Express, MySQL, Handlebars, Bootstrap, Multer.

TECHNICAL SKILLS

Languages: Kotlin, Python, JavaScript, TypeScript, PHP, SQL, C#

Mobile: Native Android, Jetpack Compose, Android Views/XML, Material 3, CameraX, Room, WorkManager, OkHttp, Coil, ZXing, React Native, Expo

Backend: FastAPI, Flask, Node.js, Express, WebSockets, REST APIs

AI / ML: TensorFlow, Keras, PyTorch, Transformers, EasyOCR, YOLOv8, SMPLer-X, Gemini API, Together AI

Databases / Cloud: Firestore, Firebase Realtime Database, Firebase Storage, MySQL, SQLite

Architecture & Methodologies: Agile / Scrum, Kanban, MVC, MVVM, Offline-first architecture, REST API design

Tools: Git, GitHub, Jira, Figma, Postman, Android Studio, VS Code, Linux/CLI, SSH

LANGUAGES

Spanish: Native

English: B2 – Professional working proficiency